4 - The Evolution of Large Scale Structure

$$\delta_k'' + \mathcal{H}\delta_k' + \left(c_s^2 k^2 - 4\pi G a^2 \bar{\rho}\right)\delta_k = 0$$

NB: Damped harmonic oscillator equation.

The physical wavelenght λ_J defines the transition between pressure dominated ($c_s^2 k^2 > 4\pi G a^2 \bar{\rho}$) and gravity dominated regimes.

$$\lambda_J = \frac{2\pi}{k_J} = c_s\left(\frac{\pi}{G\bar{\rho}}\right)^{1/2}$$

Note: $c_s^2 \sim \frac{k_B T}{m} \longrightarrow \lambda_J = \sqrt{\frac{\pi k_B T}{G m \bar{\rho}}}$

4.1 Pressureless fluid in Matter dominated era

$c_s^2 \simeq 0$ so $\lambda \gg \lambda_J$.

matter dominated era: other energy densities are negligible in front of that of matter: $\bar{\rho} \simeq \bar{\rho}_M$

$\Omega_M[\eta] = 1 \longrightarrow \rho_M = \rho_c = \frac{3\mathcal{H}^2}{8\pi G a^2}$

$$\delta''_k + \mathcal{H}\delta'_k - \frac{3}{2}\mathcal{H}^2\delta_k = 0$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \left(\frac{\dot{a}}{a}\right)_0^2 \left(\Omega_M \left(\frac{a_0}{a}\right)^3\right) \rightarrow \text{conformal.}\quad a\,d\eta = dt$$

$$\hookrightarrow \quad a \propto \eta^2 \quad \left(\text{separating var } \eta \text{ \& } a \text{ and integrating.}\right.$$

so $\mathcal{H} = \frac{2}{\eta}$ So try power law sols $\delta_k \propto \eta^q$

$$\rightarrow \quad \delta_k = C_1 \eta^2 + C_2 \eta^{-3}$$

where C_1, C_2 are constants of integration.

so $\underbrace{\delta_k \propto a}$ & $\underbrace{\delta_k \propto a^{-3/2}}$

growing mode decaying mode

in terms of "t" $\rightarrow$ $\delta_k \propto t^{2/3}$ & $\delta_k \propto 1/t$

4.2. Pressureless Matter in Λ dominated era

$c_s^2 \simeq 0$ and $\lambda \gg \lambda_J$ However $\bar{\varepsilon} \simeq \bar{\varepsilon}_r$

$$\Omega_r[\eta] = 1 \rightarrow \varepsilon_r[\eta] = \varepsilon_c = \frac{3\mathcal{H}^2}{8\pi a^2 G}$$

No fluctuation in ρ_r to source grav. pot (by definition) i.e. No term $4\pi G \rho_r \delta_r$ with $\delta_r \neq 0$ in Newton-Poisson eq.

$$\delta_k'' + \mathcal{H}\delta_k' - 4\pi G a^2 \rho_M \delta_k$$

$$= \delta_k'' + \mathcal{H}\delta_k' - \frac{3}{2}\mathcal{H}^2 \frac{\rho_M}{\rho_r}\delta_k = 0$$

$$\rho_M \ll \rho_r = \bar{\rho}$$

get $a \propto \frac{1}{\eta}$ so $\mathcal{H} = -1/\eta$ try power law solutions: $\delta_k \propto \eta^\alpha$

$$\rightarrow \delta_k = C_1 + C_2 \eta^2$$

or $\delta_k = C^{st}$ and $\delta_k \propto a^{-2}$ → decays

↳ "growing" term does not grow!

For normal time "t" $\rightarrow \delta_k = C^{st}$ and

$$\delta_k \propto \exp\left(-2\sqrt{\Lambda/3}\, t\right)$$

$$a \propto \exp\left(\sqrt{\Lambda/3}\, t\right)$$

4.3 - Relativistic Fluid in Radiation dominated era

Radiation interacting strongly with baryons before matter equality. Radiation behaves as a gravitating fluid with pressure $\bar{P} = \frac{1}{3}\bar{\rho}c^2$

$c_s = \frac{1}{\sqrt{3}} c$.

We could extend the Newtonian approach to deal with this case. Treat the fluid as special-relat and sourcing the potential. In this case:

- $\dot{\delta} + \frac{4}{3a} \nabla \vec{v} = 0$
- $\dot{\vec{v}} + H\vec{v} = -\frac{\nabla\phi}{a} - \frac{c^2 \nabla\delta}{3a}$
- $\nabla^2\phi = 8\pi G \bar{\rho} a^2 \delta$

$\bar{\rho} \approx \rho_\gamma \rightarrow \Omega_\gamma[\eta] = 1$

$\rho_\gamma[\eta] = \rho_c[\eta] = \frac{3\mathcal{H}^2}{8\pi a^2 G}$

$a \propto \eta$ so $\mathcal{H} = 1/\eta$

$\mathcal{H}^2 = -\mathcal{H}' = 1/\eta^2$

$\phi'' + 3\mathcal{H}\phi' + (2\mathcal{H}' + \mathcal{H}^2)\phi - 4\pi G a^2 \bar{\rho} \delta c_s^2 = 0$

rewrite Newton-Poisson relat. eq.

$$4\pi G a^2 \bar{\rho}\delta = \nabla^2\phi - 3\frac{\mathcal{H}}{c^2}(\mathcal{H}\phi + \phi')$$

get rid of δ term & Fourier transform

$$\phi_k'' + 4\mathcal{H}\phi_k' + c_s^2 k^2 \phi_k = 0$$

$$\longrightarrow \phi_k'' + \frac{4}{\eta}\phi_k' + \frac{c^2}{3}k^2\phi_k = 0$$

Solve:

$$\phi_k = A_1 \frac{\sqrt{3}}{kc\eta} j_1\left(\frac{kc\eta}{\sqrt{3}}\right) + A_2 \frac{\sqrt{3}}{kc\eta} y_1\left(\frac{kc\eta}{\sqrt{3}}\right)$$

$$j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}$$

$$y_1(x) = -\frac{\cos x}{x^2} - \frac{\sin x}{x}$$

Early times : $\eta \longrightarrow 0 \Rightarrow A_2 = 0$

If $kc\eta \gg 1$

$$\phi_k \simeq -\frac{3A_1}{(kc\eta)^2}\cos\left(\frac{kc\eta}{\sqrt{3}}\right)$$

$$\delta_k = \frac{2}{3\mathcal{H}^2}\left(-k^2\phi_k - 3\frac{\mathcal{H}}{c^2}(\mathcal{H}\phi_k + \phi_k')\right)$$

$$= -\left(\frac{2}{3}k^2\eta^2 + \frac{2}{c^2}\right)\phi_k - \frac{2\eta}{c^2}\phi_k'$$

for $kc\eta \gg 1$ pert in sol for ϕ_k.

$$\delta_k = -\frac{2}{3}k^2\eta^2\phi_k = \frac{2A_1}{c^2}\cos\left(\frac{kc\eta}{\sqrt{3}}\right)$$

oscillate around $\delta_k = 0$ @ freq $\frac{kc}{\sqrt{3}}$

4_4 Pressureless matter in radiation dominated era

$c_s^2 \simeq 0$. $\rho_\gamma[\eta] = \rho_c[\eta] = \frac{3\eta^2}{8\pi G a^2}$

ex: CDM during rad. dominated era for $kc\eta \gg 1$

$$\nabla^2\phi = 4\pi G(\rho_\gamma\delta_\gamma + \rho_M\delta_M)$$

$$= 4\pi G\rho_\gamma\left(\delta_\gamma + \frac{\rho_M}{\rho_\gamma}\delta_M\right)$$

$$\simeq 4\pi G\rho_\gamma\delta_\gamma$$

$$\langle\delta_\gamma\rangle_t = 0$$

so $\delta_k'' + \mathcal{H}\delta_k' = \delta_k'' + \frac{1}{\eta}\delta_k' = 0$

sol: $\delta_k = C_1\ln\eta + C_2$

Contrary to $\delta_{\gamma k}$ which does not grow,

$\delta_{k,\mathrm{CDM}}$ grows as $\ln a$.